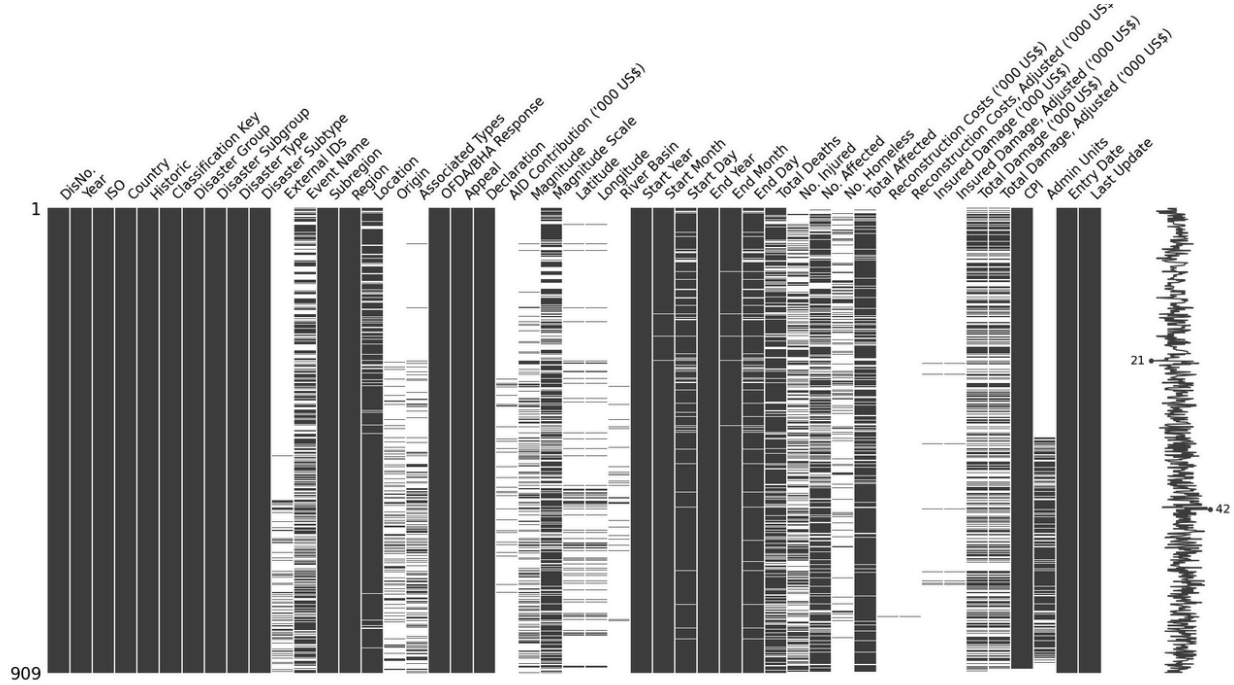
In this analysis – we had brought in two datasets: EM_DAT disaster data and WDI, world economic data from 1960 - 2023/24.

Disaster Data

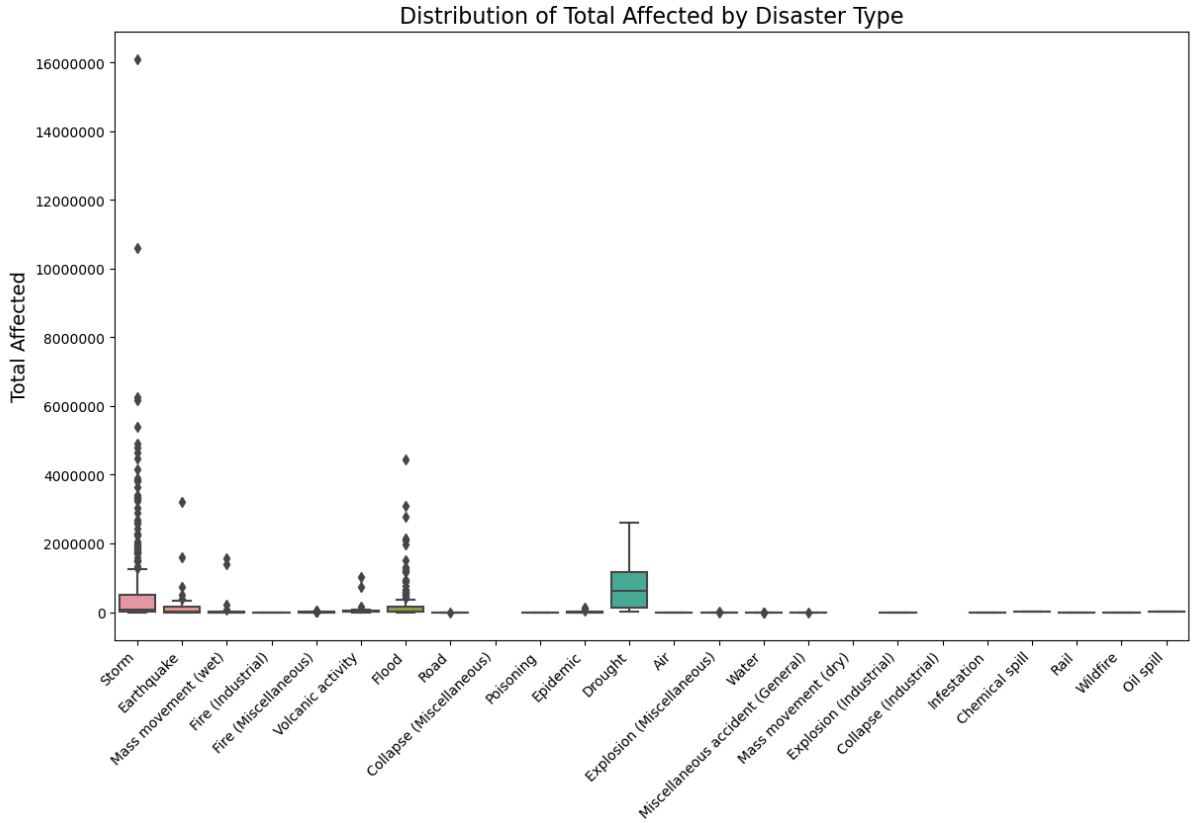
Within philip_EDA, I had initially viewed the data and realized much of the disaster column values were missing/ NULL values, using “missingno” to visualize this:



: white indicates missing values within the column space

For the most filled-in disaster indicator, I decided to use “Total Affected” to give magnitude as to which given disasters had the most impact/ severity on the community. In order to weed out weaker storms that would not have much impact on the economy/community in general.

I then wanted to visualize a “top” amount of storms that may be out of the ordinary and most impactful, and to decide which Total Affected number would be my cut-off.
I used a box-plot to view the Total Affected distribution across Disaster Types:



It wasn't surprising that "Storms", comprising of Tropical cycles/Typhoons, and "Floods" were the most impactful in the Philippines. I decided to use 2mil as a cut-off amount for Total Affected and decided all storms with 2mil or more Total Affected would be graphed from 1960 to 2023/24.

This box-plot representation also led me to look into the outlier of a storm affecting 16mil (according to the EM_DAT data value) and led me to read about Typhoon Haiyan which had a devastating earthquake occur just a month before.
According to
https://www.unicefusa.org/what-unicef-does/emergency-response/weather-disasters/typhoons/2013-philippines-typhoon-haiyan

Disaster in the Philippines

Super Typhoon Haiyan, also known as Super Typhoon Yolanda, was one of the most powerful tropical storms on record. With winds reaching 195 miles per hour, the typhoon caused tremendous damage when it made landfall in the Philippines on November 8, 2013 — less than a month after a 7.2 magnitude earthquake struck Bohol — affecting over 14 million Filipinos, including nearly 6 million children. More than 6,000 people were killed, and thousands of homes were destroyed.

However, this didn't show up in the EM_DAT data for Total Affected:

```
storms_2013 = disas_data[(disas_data["Year"] == 2013) & (disas_data["End Month"] >= 10.0)]
storms_2013[["Disaster Type", "Disaster Subtype", "End Year", "End Month", "Total Affected"]]
```

	Disaster Type	Disaster Subtype	End Year	End Month	Total Affected
733	Earthquake	Ground movement	2013	10.0	3222224.0
735	Storm	Tropical cyclone	2013	10.0	871755.0
736	Storm	Tropical cyclone	2013	11.0	16106870.0
737	Flood	Riverine flood	2013	10.0	207971.0
738	Storm	Tropical cyclone	2013	10.0	220443.0
739	Road	Road	2013	12.0	20.0

:Where it shows as 3mil Total Affected.

Despite this, the **year 2013** may show significant changes and trends within the economic data for the Philippines (according to our initial project objective).

—

Further looking into the disaster metrics, I did graph these storms – however, I later read from another academic source that top 1% most impactful disasters had an affect on the economy and even top 5% significantly less impact (source: The Economic Impacts of Natural Disasters: A Review of Models and Empirical Studies, Boltzen 2019). As stated, “a disaster in the top 1 percent of the disaster index distribution reduces the GDP growth rate by 7 percent, while a disaster in the top 5 percent of the distribution reduces it by only 0.5 percent” – and to keep the project design easier to visually comprehend, I limited my disasters to the top 1% based on Total Affected.

```
# 99% data
threshold = disas_data['Total Affected'].quantile(0.99)

# top 1%
top_1percent_disasters = disas_data[disas_data['Total Affected'] >= threshold]
print(top_1percent_disasters.shape)
top_1percent_disasters[["Disaster Subtype","Year","Total Affected"]]
```

```
(8, 47)
```

:

	Disaster Subtype	Year	Total Affected
305	Tropical cyclone	1990	6159569.0
588	Tropical cyclone	2008	4785460.0
630	Tropical cyclone	2009	4901763.0
721	Tropical cyclone	2012	6246664.0
736	Tropical cyclone	2013	16106870.0
742	Tropical cyclone	2014	4654966.0
867	Tropical cyclone	2021	10608996.0
890	Tropical cyclone	2023	5385237.0

Econ Indicators Data (WDI dataset)

Within the WDI dataset, I had first explored (for the 1488 attributes in Philippines) about a little less than half (~600) had most of their values present.
Values were considered present if out of the 64 years, they had over half with values (over 32 actual values).

```
how_empty = econ_data[econ_data['NaN_count'] < 32] # 64 years for econ attributes, are the years mostly empty?
how_empty.shape
```

```
(636, 69)
```

As we can see down below, the NaN count found Null values and quite a few econ indicators had more than half of their yearly data missing (numbers from 34 - 60)

```
econ_data['NaN_count'] = econ_data.iloc[:, 4:].isna().sum(axis=1) # how many values are NaN ? Create a NaN_count column for each econ attribute

print(econ_data[['Indicator Name', 'NaN_count']])
```

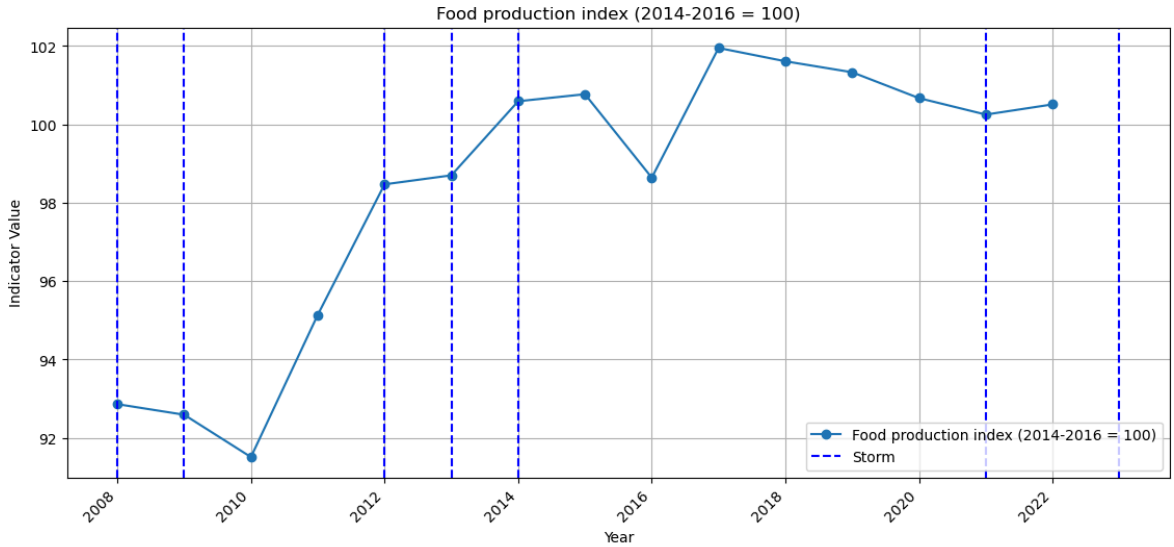
```
                                      Indicator Name  NaN_count
0     Access to clean fuels and technologies for coo...         41
1     Access to clean fuels and technologies for coo...         41
2     Access to clean fuels and technologies for coo...         41
3               Access to electricity (% of population)         34
4     Access to electricity, rural (% of rural popul...         34
...                                                 ...        ...
1483  Women who believe a husband is justified in be...         59
1484  Women who were first married by age 15 (% of w...         57
1485  Women who were first married by age 18 (% of w...         57
1486  Women's share of population ages 15+ living wi...         31
1487  Young people (ages 15-24) newly infected with HIV         31

[1488 rows x 2 columns]
```

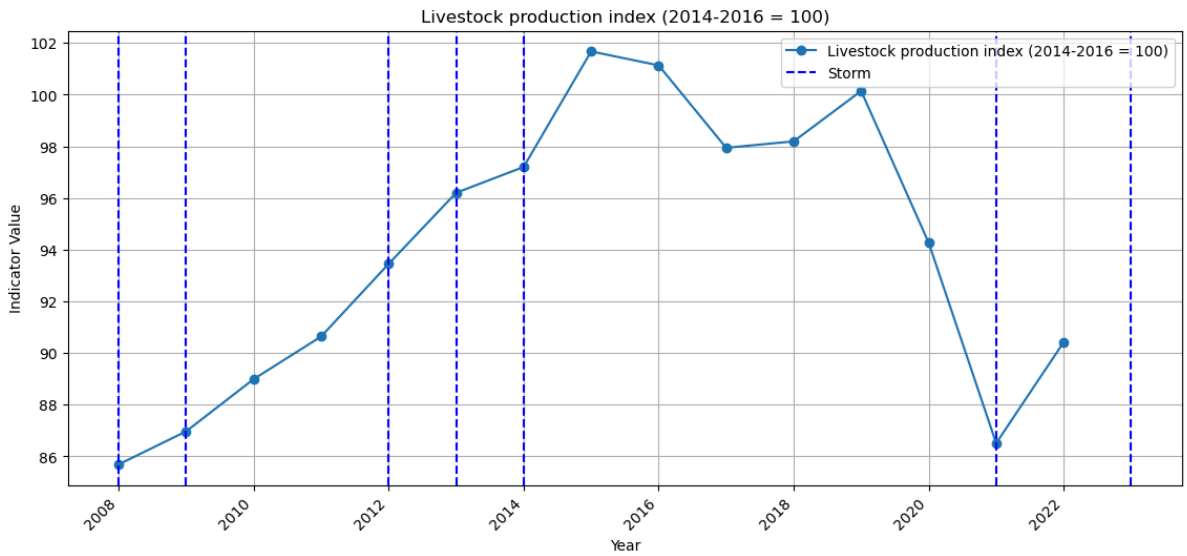
So I decided to only use econ indicators with present values more than half (greater than 32 values were non-null).

I then made an initial graph with years between 1960 - 2023/24; however, further into our project, we decided since the WDI data is more accurate for recent years, we would use the data from 2008 - 2023/24.

I first graphed the first 100 indicators, but we had decided to hone in on GDP/GNI indicators, emissions, and production indicators. Graphing between 2008 - 2023/24 (now 7 prominent storms) and focusing on these indicator groups, I did not see much trend behavior for the Philippines in this data like we had hoped/ compared to Puerto Rico and Hurricane Maria.

```
     Year  Total Affected
588  2008       4785460.0
630  2009       4901763.0
721  2012       6246664.0
736  2013      16106870.0
742  2014       4654966.0
867  2021      10608996.0
890  2023       5385237.0
```

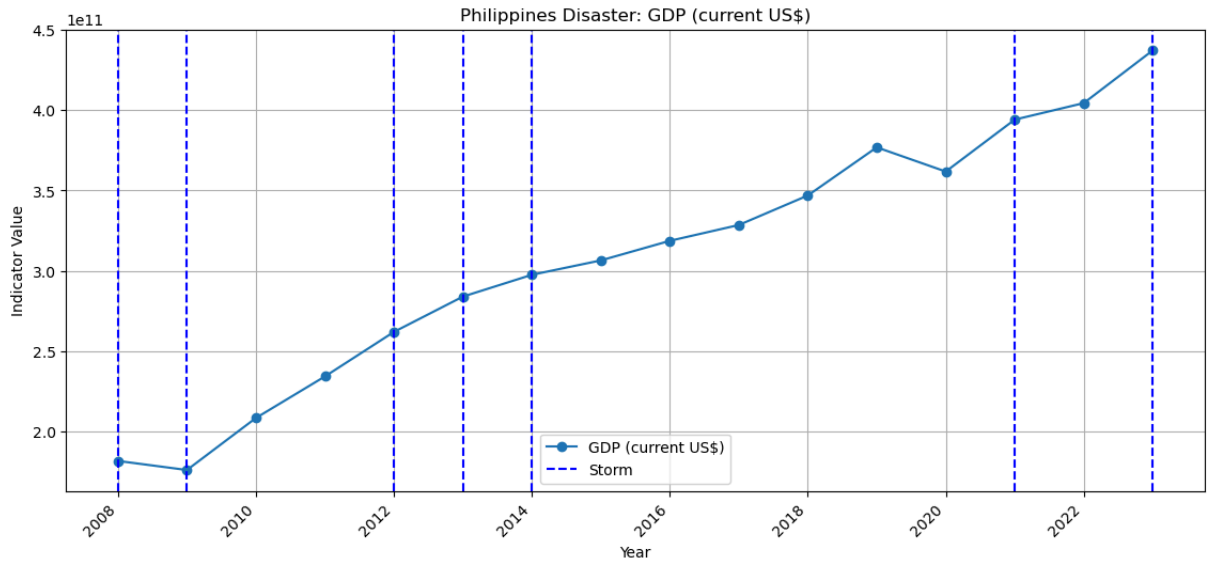


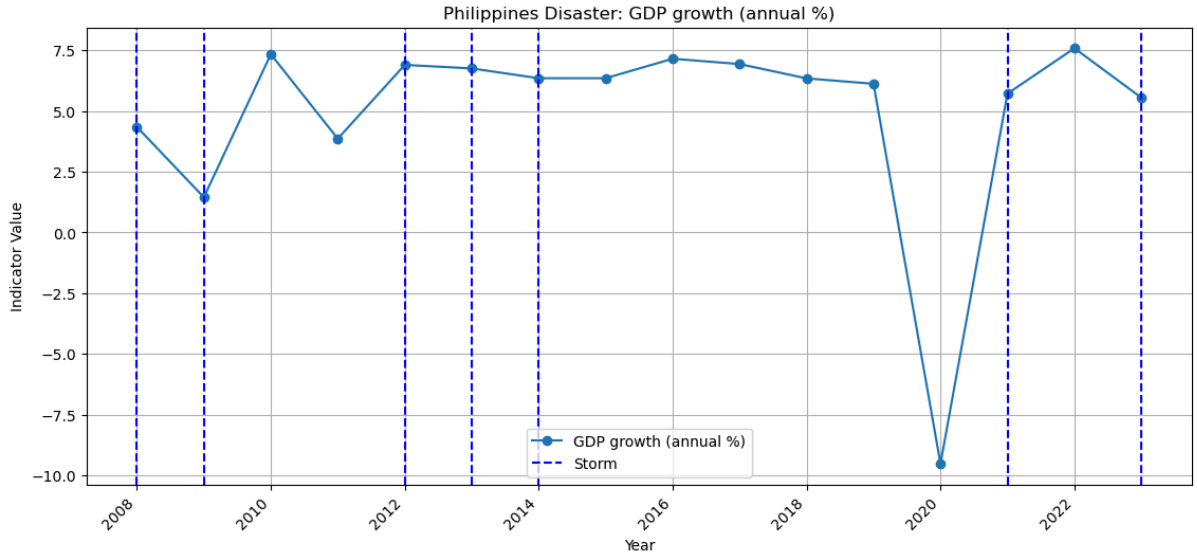


: Disasters are briefly listed with Year, Total Affected and indicated with a blue dotted line in each time-series graph.

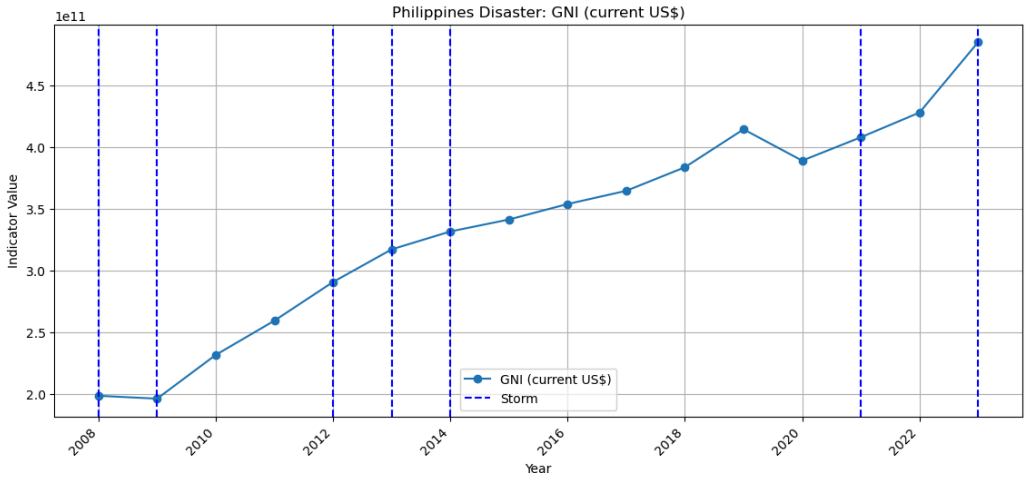
As we can see, there are no prominent downward trends and production even seems to continue to rise shortly following the Typhoon in 2013.

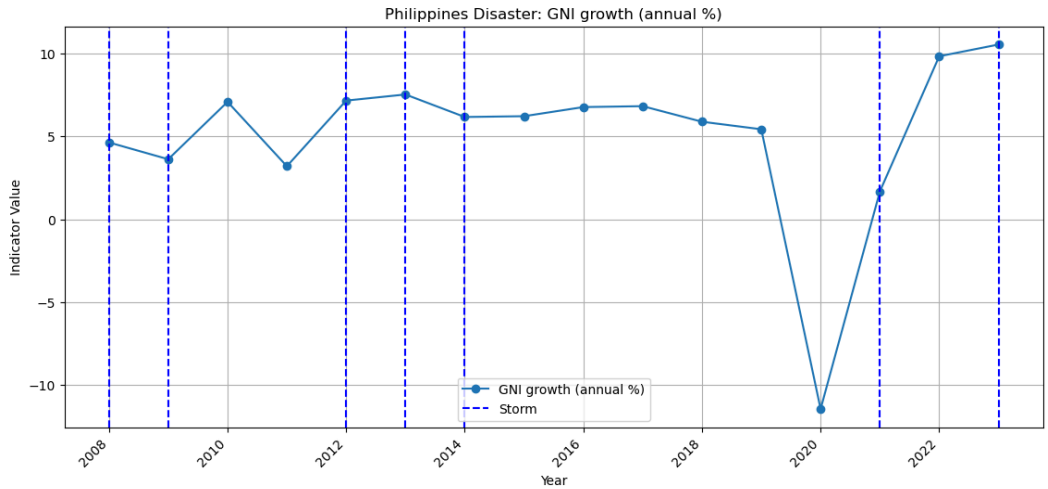
GDP/GNI attributes (more visualized within the philip_final notebook), I wanted to show that a couple of GDP metrics even show little to know effect from top 1% devastating disasters from year to year.





We can even see the drop, exaggerated in the GDP growth (annual %) indicator, for COVID in 2019/2020 as it was a global epidemic and affected most countries globally (as seen in other countries trends) but we do not see any effect after a disaster (indicated with blue dotted lines).





While GDP shows domestic production, GNI measures national income and even this indicator shows consistent trends despite disasters.

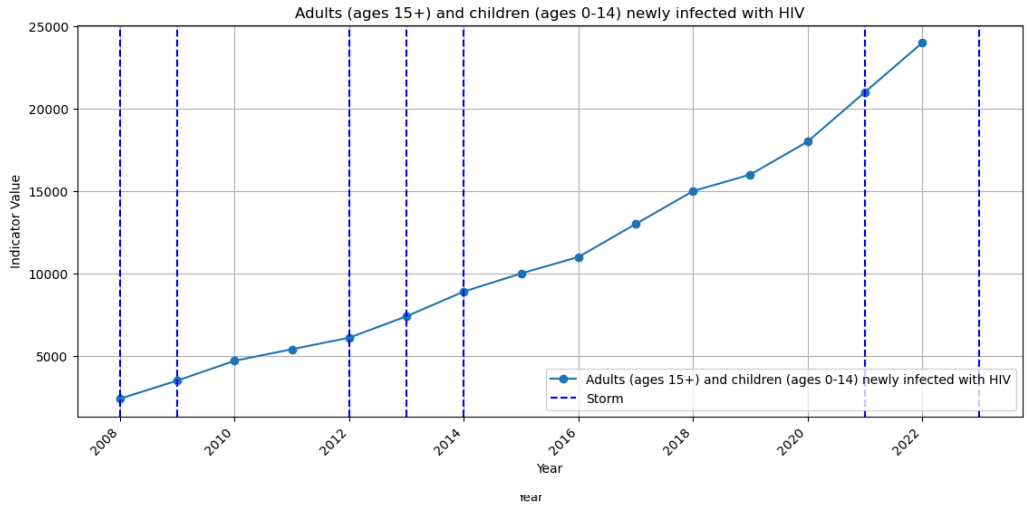
<u>Granularity Issue within our Project:</u>

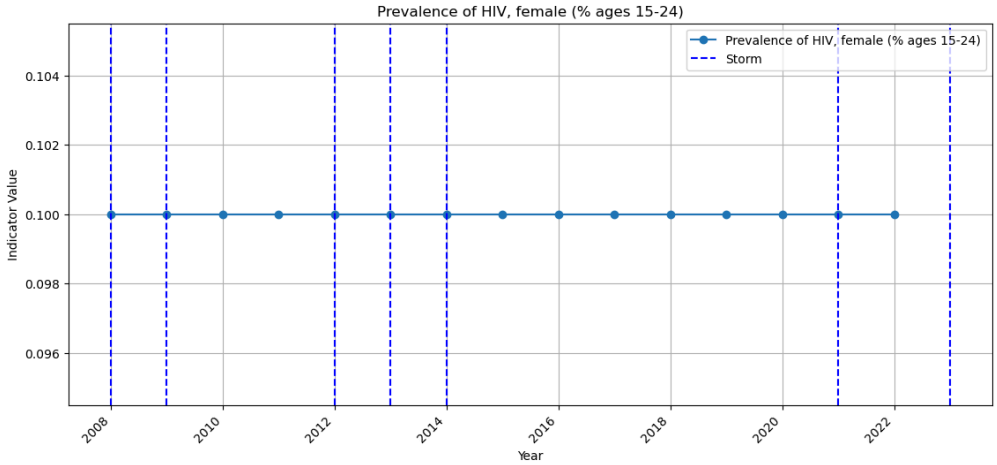
I do believe perhaps economic indicators such as this need sub-regional data vs country data and month to month vs year to year **granularity**. I believe country economies are closely affected by global economy and disasters may only greatly have an effect on specific region/sub-national locations they strike (state or province vs country).

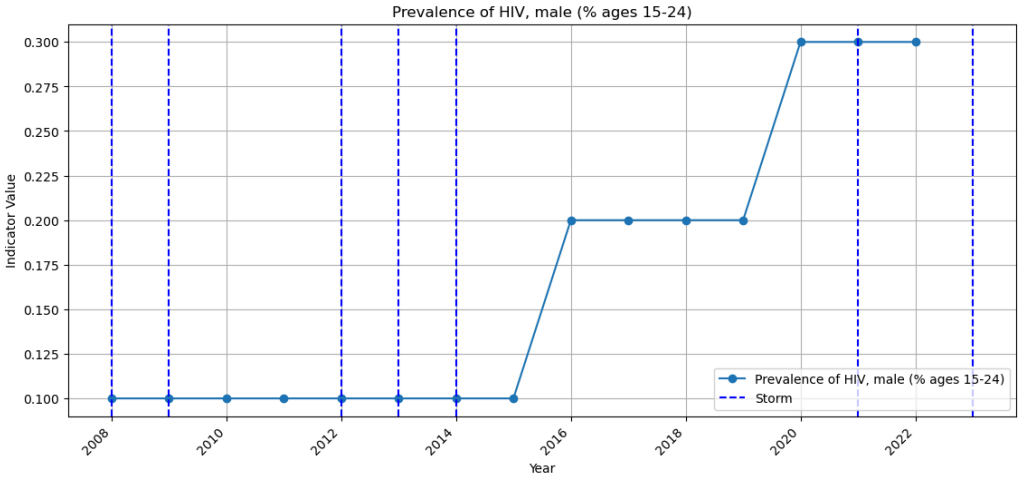
We also may have been able to see small changes in country data if we had monthly data vs yearly. Disasters have a devastating effect on community and local economy but a country is able to bounce back with recovery assistance, government funding and a stabilized economy within nearby regions in the country that were not hit.

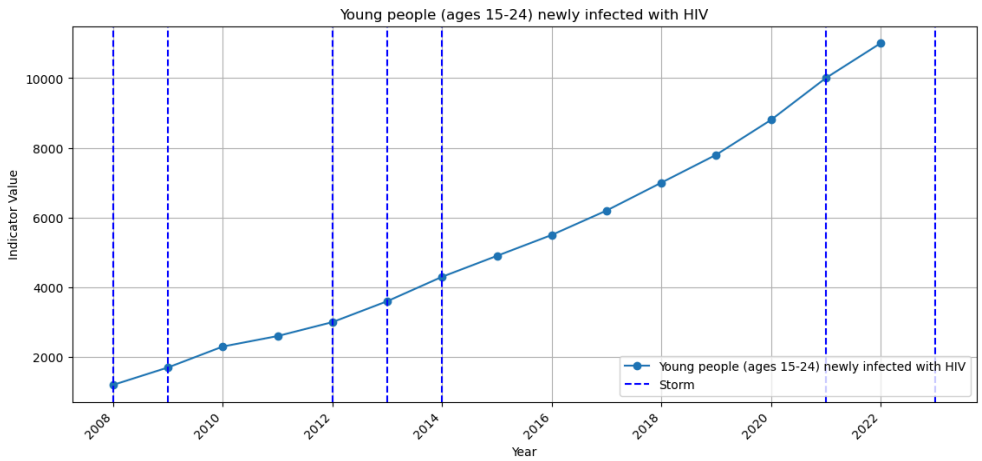
Amongst our members, we did initially speak of granularity/scope perhaps changing to include regional/sub-location data – however, viewing every indicator within the WDI dataset took most of our time before we were able to find, clean and merge multiple economic datasets to showcase regional data and the WDI does not separate their metrics into smaller regions. We chose to spend most of our time working within WDI.

Later, I wanted to graph HIV metrics, just to see if population indicators could be affected by an impactful disaster.





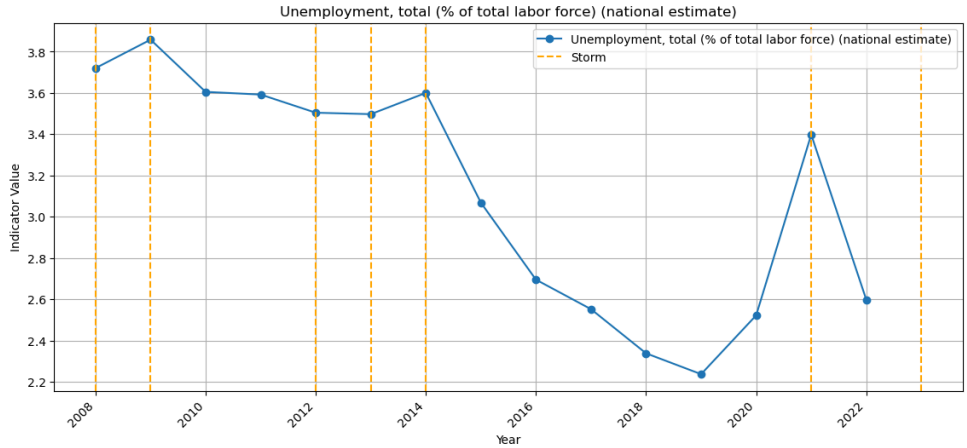


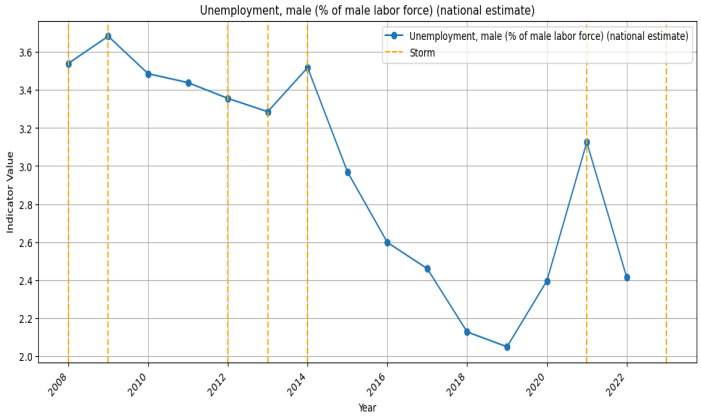


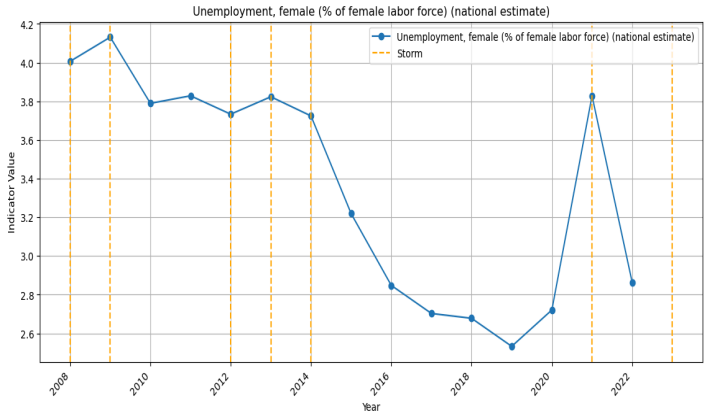
As we can see, even within Female to Male and Youth demographics, disasters had no national effect on the Philippines population – and they trends upwards dependent.

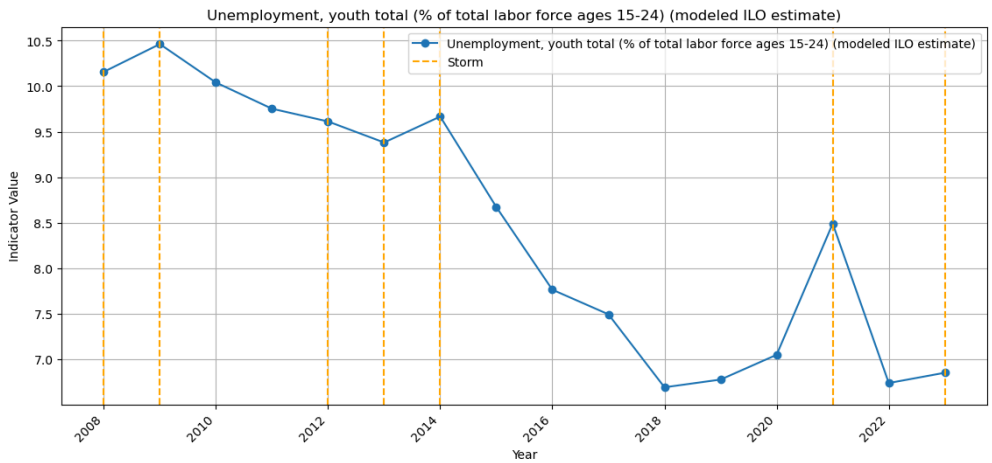
Lastly, I did find an indicator group that seemed to be affected by disasters and that was Unemployment.

We see that in all demographics (Male vs Female and Youth) and total population, unemployment trends downward after top 1% Total Affected storms.









Overall, the Typhoon in 2013 (and other storms in 2013) didn't seem to have an effect in most yearly economic indicators. One could do more research into why, country-specific reasons or global economic reasons – maybe disaster relief action was sufficient during this time and we can learn something from the Philippine's actions.

<subsequently, I went through a similar process with Japan and China and found no significant trends>